

100 Most Important Syndromes

High-Yield Buzzwords & Clues for FMGE & NEET-PG

1. Down Syndrome (Trisomy 21)

- Single palmar (simian) crease
- Sandal gap between 1st & 2nd toes
- Epicanthal folds & flat facial profile
- Associated with VSD/AV canal defect
- Increased risk of Alzheimer's disease

2. Turner Syndrome (45, XO)

- Webbed neck (cystic hygroma in utero)
- Streak ovaries (primary amenorrhea)
- Coarctation of the aorta
- Bicuspid aortic valve
- Horseshoe kidney

3. Klinefelter Syndrome (47, XXY)

- Tall stature with long extremities
- Gynecomastia & testicular atrophy
- Hypergonadotropic hypogonadism
- Barr body positive in males
- Eunuchoid body shape

4. Marfan Syndrome

- Fibrillin-1 gene mutation (chromosome 15)
- Upward/superotemporal lens subluxation
- Aortic root dilation / dissection
- Arachnodactyly & pectus excavatum
- Mitral valve prolapse

5. Ehlers-Danlos Syndrome

- Collagen synthesis defect (Type I/III/V)
- Hyperextensible skin & hypermobile joints
- Berry aneurysms (vascular type)
- Poor wound healing
- Joint dislocations easily

6. Fragile X Syndrome

- CGG trinucleotide repeat (FMR1 gene)
- Macroorchidism (enlarged testes)
- Long face with large jaw and everted ears
- Most common inherited cause of intellectual disability
- Mitral valve prolapse

7. Patau Syndrome (Trisomy 13)

- Cleft lip/palate
- Microphthalmia (small eyes)
- Polydactyly
- Holoprosencephaly
- Cutis aplasia (scalp defects)

8. Edwards Syndrome (Trisomy 18)

- Prominent occiput
- Rocker-bottom feet
- Intellectual disability
- Clenched fists with overlapping fingers
- Micrognathia

9. DiGeorge Syndrome

- 22q11 microdeletion
- Thymic, Parathyroid, and Cardiac defects
- Hypocalcemia (tetany/seizures)
- Conotruncal cardiac anomalies (Tetralogy of Fallot)
- Cleft palate

10. Prader-Willi Syndrome

- Maternal imprinting (Paternal 15q deletion)
- Hyperphagia leading to obesity
- Hypogonadism & hypotonia
- Almond-shaped eyes
- Behavioral problems

11. Angelman Syndrome

- Paternal imprinting (Maternal 15q deletion)
- Happy puppet (inappropriate laughter)
- Severe intellectual disability
- Seizures & ataxia
- UBE3A gene mutation

12. Rett Syndrome

- MECP2 gene mutation (X-linked dominant)
- Regression of motor/verbal skills in girls
- Hand-wringing stereotypy
- Normal development until 6-18 months
- Lethal in males

13. Williams Syndrome

- Microdeletion on chromosome 7q11.23
- Elfin facies
- Cocktail party personality (extremely friendly)
- Supravalvular aortic stenosis
- Hypercalcemia in infancy

14. Cri-du-chat Syndrome

- Microdeletion of chromosome 5p
- High-pitched, cat-like cry
- Microcephaly
- Epicanthal folds
- Severe intellectual disability

15. Sturge-Weber Syndrome

- GNAQ gene mutation
- Port-wine stain (Nevus flammeus) in V1/V2 distribution
- Ipsilateral leptomeningeal angioma
- Seizures/epilepsy
- Early onset glaucoma

16. Tuberous Sclerosis

- TSC1 (hamartin) / TSC2 (tuberin) mutation
- Ash-leaf spots & Shagreen patches
- Cardiac rhabdomyomas
- Subependymal giant cell astrocytomas
- Renal angiomyolipomas

17. Neurofibromatosis Type 1

- NF1 gene mutation (chromosome 17)
- Café-au-lait spots (>6)
- Lisch nodules (iris hamartomas)
- Optic gliomas
- Multiple neurofibromas

18. Neurofibromatosis Type 2

- NF2 gene mutation (chromosome 22)
- Bilateral acoustic neuromas (schwannomas)
- Meningiomas
- Juvenile cataracts
- Ependymomas

19. Von Hippel-Lindau Disease

- VHL gene mutation (chromosome 3)
- Hemangioblastomas (retina, brain stem, cerebellum)
- Bilateral Renal Cell Carcinomas (clear cell)
- Pheochromocytomas
- Pancreatic cysts

20. Peutz-Jeghers Syndrome

- STK11 mutation
- Hamartomatous GI polyps
- Hyperpigmented macules on lips/oral mucosa
- Increased risk of breast and GI cancers
- Intussusception risk

21. Familial Adenomatous Polyposis (FAP)

- APC gene mutation (chromosome 5q)
- Thousands of colonic polyps
- 100% risk of colorectal cancer
- Prophylactic colectomy required
- Pancolonic involvement

22. Gardner Syndrome

- Variant of FAP
- Osteomas (often of the mandible/skull)
- Epidermoid cysts & fibromatosis
- Supernumerary teeth
- High malignant potential

23. Turcot Syndrome

- Variant of FAP or Lynch Syndrome
- GI polyposis
- Malignant CNS tumors
- Medulloblastoma
- Glioblastoma multiforme

24. Lynch Syndrome (HNPCC)

- Mismatch repair gene mutation (MLH1, MSH2)
- Colorectal cancer (often right-sided) without polyposis
- Endometrial and ovarian cancers
- Microsatellite instability
- Skin cancers

25. Zollinger-Ellison Syndrome

- Gastrin-secreting tumor (gastrinoma)
- Multiple refractory peptic ulcers
- Jejunal ulcers (distal to duodenum)
- Associated with MEN1 syndrome
- Secretin stimulation test positive

26. Budd-Chiari Syndrome

- Hepatic vein thrombosis
- Hepatomegaly & Ascites
- Abdominal pain
- Absent JVD (distinguishes from right heart failure)
- Associated with polycythemia vera

27. Crigler-Najjar Syndrome

- Absent UDP-glucuronosyltransferase (UGT)
- Severe unconjugated hyperbilirubinemia
- Kernicterus in neonates
- Often fatal (Type I) without liver transplant
- Plasmapheresis/phototherapy used

28. Gilbert Syndrome

- Mild decrease in UGT activity
- Mild, benign unconjugated hyperbilirubinemia
- Jaundice triggered by stress, fasting, or illness
- No treatment required
- Very common

29. Dubin-Johnson Syndrome

- Defective hepatic bilirubin excretion
- Conjugated hyperbilirubinemia
- Grossly black liver
- Benign condition
- Epinephrine metabolites in urine

30. Rotor Syndrome

- Similar to Dubin-Johnson but milder
- Conjugated hyperbilirubinemia
- Liver is normal color (not black)
- Benign, no treatment needed
- Defect in hepatic uptake/storage

31. Mallory-Weiss Syndrome

- Longitudinal mucosal lacerations at GE junction
- Associated with severe retching/vomiting
- Alcoholics and bulimics
- Presents with painful hematemesis
- Usually heals spontaneously

32. Boerhaave Syndrome

- Transmural, full-thickness esophageal rupture
- Due to violent vomiting
- Subcutaneous emphysema (crepitus)
- Left-sided pleural effusion with high amylase
- Surgical emergency

33. Plummer-Vinson Syndrome

- Dysphagia (esophageal webs)
- Iron deficiency anemia
- Glossitis (beefy red tongue)
- Increased risk of squamous cell carcinoma of esophagus
- Koilonychia

34. Cushing Syndrome

- Excess cortisol
- Moon facies & Buffalo hump
- Central obesity & purple striae
- Hypertension & hyperglycemia
- Proximal muscle weakness

35. Conn Syndrome

- Primary hyperaldosteronism (aldosterone-producing adenoma)
- Hypertension
- Hypokalemia
- Metabolic alkalosis
- Low renin levels

36. Addison Disease

- Primary adrenal insufficiency
- Hyperpigmentation (increased ACTH/MSH)
- Hypotension & hyperkalemia
- Fatigue, weight loss, hyponatremia
- Autoimmune destruction most common

37. Waterhouse-Friderichsen Syndrome

- Bilateral adrenal hemorrhage
- Associated with *Neisseria meningitidis* septicemia
- DIC & endotoxic shock
- Acute primary adrenal insufficiency
- Rapidly fatal without steroids

38. Sheehan Syndrome

- Postpartum ischemic necrosis of anterior pituitary
- Failure to lactate (low prolactin)
- Absent menstruation (low FSH/LH)
- Cold intolerance & fatigue (low TSH)
- Loss of pubic hair

39. Kallmann Syndrome

- Failure of GnRH-producing neurons to migrate
- Anosmia (loss of smell)
- Hypogonadotropic hypogonadism
- Delayed puberty & primary amenorrhea
- Normal secondary sexual characteristics absent

40. Pendred Syndrome

- SLC26A4 mutation
- Sensorineural hearing loss
- Goiter (thyroid swelling)
- Defect in iodide transport
- Autosomal recessive

41. Nelson Syndrome

- Enlargement of an ACTH-secreting pituitary adenoma
- Occurs after bilateral adrenalectomy (for Cushing's)
- Hyperpigmentation
- Visual field defects (bitemporal hemianopsia)
- Headaches

42. Wernicke-Korsakoff Syndrome

- Vitamin B1 (Thiamine) deficiency
- Confusion, Ophthalmoplegia, Ataxia (Wernicke)
- Confabulation & memory loss (Korsakoff)
- Damage to mammillary bodies
- Common in chronic alcoholics

43. Brown-Séquard Syndrome

- Hemisection of the spinal cord
- Ipsilateral loss of motor function (corticospinal)
- Ipsilateral loss of vibration/proprioception (dorsal column)
- Contralateral loss of pain/temp (spinothalamic)
- UMN signs below lesion

44. Horner Syndrome

- Sympathetic chain lesion
- Ptosis (drooping eyelid)
- Miosis (constricted pupil)
- Anhidrosis (lack of sweating)
- Enophthalmos

45. Wallenberg Syndrome

- Lateral medullary syndrome (PICA stroke)
- Dysphagia, hoarseness (nucleus ambiguus)
- Decreased pain/temp on ipsilateral face & contralateral body
- Ipsilateral Horner syndrome
- Vestibular signs (vertigo, nystagmus)

46. Locked-in Syndrome

- Basilar artery infarct
- Bilateral ventral pontine lesion
- Quadriplegia & loss of voluntary facial movement
- Preserved consciousness & vertical eye movements
- EEG is normal (awake)

47. Guillain-Barré Syndrome

- Acute inflammatory demyelinating polyradiculoneuropathy
- Ascending muscle weakness/paralysis
- Absent deep tendon reflexes
- Often follows Campylobacter jejuni infection
- Albuminocytologic dissociation in CSF

48. Lambert-Eaton Myasthenic Syndrome

- Antibodies against presynaptic voltage-gated calcium channels
- Proximal muscle weakness that improves with use
- Associated with Small Cell Lung Cancer (SCLC)
- Decreased DTRs
- Autonomic symptoms (dry mouth)

49. Ramsay Hunt Syndrome

- Herpes zoster oticus
- Facial nerve (CN VII) palsy
- Vesicles in the ear canal or eardrum
- Otalgia & loss of taste on anterior 2/3 of tongue
- Tinnitus/vertigo

50. Ménière's Disease

- Endolymphatic hydrops
- Episodic vertigo
- Fluctuating sensorineural hearing loss
- Tinnitus
- Ear fullness

51. Sjögren Syndrome

- Autoimmune destruction of exocrine glands
- Xerophthalmia (dry eyes)
- Xerostomia (dry mouth)
- Anti-Ro (SSA) and Anti-La (SSB) antibodies
- Bilateral parotid enlargement

52. Reactive Arthritis (Reiter Syndrome)

- 'Can't see, can't pee, can't climb a tree'
- Conjunctivitis / Uveitis
- Urethritis
- Arthritis (asymmetric)
- Following GI (Shigella) or Chlamydia infection

53. Behcet Syndrome

- Immune complex vasculitis
- Recurrent oral aphthous ulcers
- Genital ulcers
- Uveitis
- Erythema nodosum

54. Goodpasture Syndrome

- Anti-GBM antibodies (Type IV collagen)
- Hematuria (RPGN glomerulonephritis)
- Hemoptysis (pulmonary alveolar hemorrhage)
- Linear IgG deposition on immunofluorescence
- Young adult males

55. Alport Syndrome

- Mutation in Type IV collagen (X-linked)
- Sensorineural hearing loss
- Ocular abnormalities (lens dislocation)
- Hematuria & progressive renal failure
- 'Basket-weave' appearance on EM

56. Nephrotic Syndrome (General)

- Massive proteinuria ($>3.5\text{g/day}$)
- Hypoalbuminemia & severe edema
- Hyperlipidemia & oval fat bodies
- Increased risk of thromboembolism (loss of ATIII)
- Increased infection risk

57. Nephritic Syndrome (General)

- Inflammatory damage to glomeruli
- Hematuria & dysmorphic RBCs / RBC casts
- Hypertension
- Oliguria
- Mild to moderate proteinuria

58. Bartter Syndrome

- Defect in Na-K-2Cl cotransporter (thick ascending limb)
- Similar to chronic loop diuretic use
- Hypokalemia & metabolic alkalosis
- Hypercalciuria
- Presents in childhood

59. Gitelman Syndrome

- Defect in NaCl cotransporter (distal convoluted tubule)
- Similar to chronic thiazide use
- Hypokalemia & metabolic alkalosis
- Hypocalciuria & hypomagnesemia
- More benign than Bartter

60. Liddle Syndrome

- Gain of function mutation in ENaC channels (collecting duct)
- Presents like hyperaldosteronism
- Hypertension, hypokalemia, metabolic alkalosis
- Aldosterone levels are LOW
- Treated with Amiloride

61. Fanconi Syndrome

- Generalized reabsorptive defect in PCT
- Glucosuria, aminoaciduria
- Phosphaturia & proximal renal tubular acidosis (Type 2)
- Associated with Wilson disease
- Associated with multiple myeloma

62. Hemolytic Uremic Syndrome (HUS)

- Triad: Microangiopathic hemolytic anemia, Thrombocytopenia, AKI
- Often follows E. coli O157:H7 (Shiga toxin) infection
- Schistocytes on blood smear
- Common in children
- Normal PT/PTT

63. Thrombotic Thrombocytopenic Purpura (TTP)

- ADAMTS13 deficiency
- Pentad: FAT RN (Fever, Anemia, Thrombocytopenia, Renal, Neuro)
- Schistocytes & elevated LDH
- Normal PT/PTT
- Treated with plasmapheresis

64. Bernard-Soulier Syndrome

- Defect in glycoprotein Ib (GpIb)
- Defective platelet adhesion to von Willebrand factor
- Enlarged/Giant platelets
- Mild thrombocytopenia
- Bleeding time prolonged

65. Glanzmann Thrombasthenia

- Defect in glycoprotein IIb/IIIa (GpIIb/IIIa)
- Defective platelet aggregation
- Normal platelet count & morphology
- Prolonged bleeding time
- Defective clot retraction

66. Chediak-Higashi Syndrome

- LYST gene mutation (microtubule defect)
- Partial albinism
- Recurrent pyogenic infections
- Giant granules in neutrophils
- Peripheral neuropathy

67. Wiskott-Aldrich Syndrome

- WAS gene mutation (X-linked)
- Triad: WATER (Wiskott, Aldrich, Thrombocytopenia, Eczema, Recurrent infections)
- Defective T and B cell function
- Low IgM, high IgA and IgE
- Increased risk of autoimmune disease

68. Severe Combined Immunodeficiency (SCID)

- Defective cell-mediated and humoral immunity
- ADA deficiency or IL-2R gamma chain defect
- Failure to thrive & chronic diarrhea
- Absent thymic shadow on CXR
- Severe fungal/viral/bacterial infections

69. Job Syndrome (Hyper-IgE)

- STAT3 mutation (Th17 deficiency)
- FATED: Facies (coarse), Abscesses (cold staph), Teeth (retained primary), IgE (high), Dermatologic (eczema)
- Cold (non-inflamed) staphylococcal abscesses
- Bone fractures
- High IgE levels

70. Leukocyte Adhesion Deficiency

- Defect in LFA-1 integrin (CD18)
- Delayed separation of umbilical cord (>30 days)
- Recurrent bacterial infections without pus
- Neutrophilia in blood but absent at infection site
- Impaired wound healing

71. Churg-Strauss Syndrome (EGPA)

- Eosinophilic granulomatosis with polyangiitis
- Asthma & allergic rhinitis
- Eosinophilia
- p-ANCA positive
- Neuropathy (mononeuritis multiplex)

72. Wegener Granulomatosis (GPA)

- Granulomatosis with polyangiitis
- Upper respiratory (sinusitis, saddle nose)
- Lower respiratory (hemoptysis, cavitary lesions)
- Renal (RPG glomerulonephritis)
- c-ANCA positive

73. Kawasaki Disease

- CRASH and burn (Conjunctivitis, Rash, Adenopathy, Strawberry tongue, Hands/feet scaling, Fever)
- Medium vessel vasculitis in children
- Risk of coronary artery aneurysms
- Treat with IVIG and Aspirin
- Desquamation of fingers/toes

74. Brugada Syndrome

- Autosomal dominant mutation in cardiac Na⁺ channels
- Pseudo-RBBB and ST elevation in V1-V3
- Increased risk of ventricular fibrillation
- Sudden cardiac death (often during sleep)
- Common in Asian males

75. Wolff-Parkinson-White (WPW) Syndrome

- Accessory conduction pathway (Bundle of Kent)
- Delta wave on ECG
- Short PR interval & widened QRS
- Can lead to supraventricular tachycardia
- Pre-excitation syndrome

76. Dressler Syndrome

- Autoimmune fibrinous pericarditis
- Occurs 2-10 weeks post-myocardial infarction
- Fever, pleuritic chest pain
- Friction rub heard on auscultation
- Treat with NSAIDs

77. Leriche Syndrome

- Aortoiliac occlusive disease
- Triad: Claudication (buttock/thigh), Absent/decreased femoral pulses, Impotence
- Atherosclerosis of abdominal aorta at bifurcation
- Diminished lower extremity pulses
- Muscle atrophy

78. Eisenmenger Syndrome

- Uncorrected left-to-right shunt (VSD, ASD, PDA)
- Pulmonary hypertension causes shunt reversal (right-to-left)
- Late-onset cyanosis
- Clubbing
- Polycythemia

79. Kartagener Syndrome

- Primary Ciliary Dyskinesia (Dynein arm defect)
- Situs inversus (dextrocardia)
- Chronic sinusitis
- Bronchiectasis
- Infertility (immotile sperm/dysfunctional fallopian tubes)

80. Acute Respiratory Distress Syndrome (ARDS)

- Non-cardiogenic pulmonary edema
- Bilateral widespread pulmonary infiltrates (white out)
- Normal PCWP (<18 mmHg)
- Severe hypoxemia refractory to oxygen therapy
- Hyaline membrane formation

81. Superior Vena Cava (SVC) Syndrome

- Obstruction of SVC (often due to bronchogenic carcinoma)
- Facial swelling and plethora
- Jugular venous distention
- Dilated collateral veins on upper chest
- Worse when leaning forward

82. Pancoast Syndrome

- Apical lung tumor (usually squamous cell or adenocarcinoma)
- Ipsilateral Horner syndrome
- Shoulder pain radiating to arm (C8-T1)
- Hoarseness (recurrent laryngeal nerve)
- SVC syndrome risk

83. Meigs Syndrome

- Triad: Ovarian fibroma, Ascites, Right-sided pleural effusion
- Resolves with tumor removal
- Benign condition despite presentation
- Solid adnexal mass
- Pulling sensation in groin

84. Fitz-Hugh-Curtis Syndrome

- Perihepatitis associated with PID
- Violin string adhesions between liver and peritoneum
- RUQ pain exacerbated by breathing
- Associated with Chlamydia or Gonorrhea
- Normal LFTs

85. Asherman Syndrome

- Intrauterine adhesions/synechiae
- Secondary amenorrhea
- Often history of D&C or severe pelvic infection
- Diagnosis via hysteroscopy
- Infertility

86. HELLP Syndrome

- Hemolysis, Elevated Liver enzymes, Low Platelets
- Severe form of preeclampsia
- Schistocytes on blood smear
- Right upper quadrant pain (hepatic capsule distention)
- Requires prompt delivery

87. Polycystic Ovary Syndrome (PCOS)

- Hyperandrogenism (hirsutism, acne)
- Oligo/anovulation (irregular menses)
- Polycystic ovaries on ultrasound
- Insulin resistance & obesity
- Increased LH/FSH ratio

88. Swyer Syndrome

- 46, XY pure gonadal dysgenesis
- Female phenotype but lacks functional ovaries (streak gonads)
- No pubertal development (primary amenorrhea)
- Risk of dysgerminoma/gonadoblastoma
- Normal vagina and uterus

89. Androgen Insensitivity Syndrome (Morris)

- 46, XY with mutated androgen receptor
- Female external genitalia with blind vaginal pouch
- Absent uterus and fallopian tubes
- Cryptorchid testes present (must be removed)
- Normal breast development

90. Reye Syndrome

- Hepatic encephalopathy in children
- Follows viral illness (VZV, Influenza) treated with Aspirin
- Microvesicular fatty change in liver
- Vomiting, confusion, seizures, coma
- Hypoglycemia

91. Stevens-Johnson Syndrome

- Severe mucocutaneous reaction (<10% body surface area)
- Often triggered by drugs (Sulfa, Lamotrigine, Allopurinol)
- Targetoid skin lesions and mucosal ulceration
- Fever and bullae formation
- Positive Nikolsky sign

92. Toxic Shock Syndrome

- Staph aureus exotoxin (TSST-1) superantigen
- Fever, hypotension
- Diffuse erythroderma (sunburn-like rash)
- Desquamation of palms and soles
- Often linked to prolonged tampon use

93. Staphylococcal Scalded Skin Syndrome

- Exfoliative toxins (ETA, ETB) from Staph aureus
- Cleaves desmoglein-1 in stratum granulosum
- Sloughing of skin with positive Nikolsky sign
- No mucosal involvement (unlike SJS)
- Heals without scarring

94. Carpal Tunnel Syndrome

- Median nerve compression at the wrist
- Pain/numbness in first 3.5 digits
- Thenar atrophy
- Positive Tinel and Phalen signs
- Worse at night

95. Thoracic Outlet Syndrome

- Compression of lower brachial plexus and subclavian vessels
- Often due to cervical rib or scalene muscle hypertrophy
- Atrophy of intrinsic hand muscles
- Ischemia and pain in upper extremity
- Positive Adson test

96. Compartment Syndrome

- Increased pressure within a fascial space
- Pain out of proportion to injury
- 6 Ps (Pain, Pallor, Pulselessness, Paresthesia, Poikilothermia, Paralysis)
- Requires urgent fasciotomy
- Common after long bone fractures

97. Cauda Equina Syndrome

- Compression of spinal nerve roots below L2
- Saddle anesthesia
- Bowel/bladder incontinence (late sign)
- Asymmetric lower extremity weakness & decreased DTRs
- Surgical emergency

98. Conus Medullaris Syndrome

- Lesion of sacral spinal cord (T12-L2 level)
- Early, severe bowel and bladder dysfunction
- Symmetric saddle anesthesia
- Upper and lower motor neuron signs
- Impotence

99. Tumor Lysis Syndrome

- Massive cell death after chemotherapy
- Hyperkalemia, Hyperphosphatemia, Hyperuricemia
- Hypocalcemia
- Acute kidney injury (urate or calcium phosphate crystals)
- Prevent with Allopurinol/Rasburicase

100. Carcinoid Syndrome

- Secretion of serotonin by neuroendocrine tumors
- Flushing, diarrhea, and wheezing
- Right-sided valvular heart disease (TR, PS)
- Elevated 5-HIAA in urine
- Niacin deficiency (Pellagra) risk